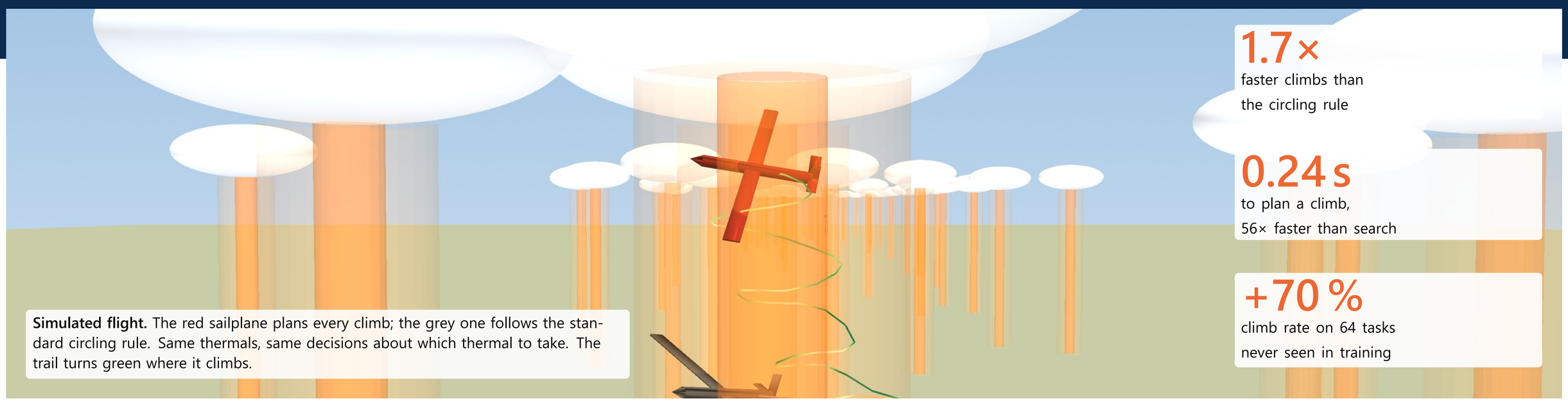
GeneticWing

A sailplane that learns how to climb in thermals

Clément Dumeril | simulation study, 2026 | clementdumeril.github.io/GeneticWing

fly it in 3D in your browser



Simulated flight. The red sailplane plans every climb; the grey one follows the standard circling rule. Same thermals, same decisions about which thermal to take. The trail turns green where it climbs.

How it works

every candidate is flown in the simulator before it is kept: no plan breaks the aircraft's limits

Teacher — a slow optimizer solves **20,923** single-thermal climbs → **Command VAE** — learns them; writes a climb as **16 numbers**, flown by a simulator → **Genetic search** — searches the 16 numbers; beats the teacher but takes **13 s** → **Generator** — flow matching learns the search's answers: 64 ideas, best kept, **0.24 s** → **Cross-country** — a tactician picks the thermal, the generator flies the climb

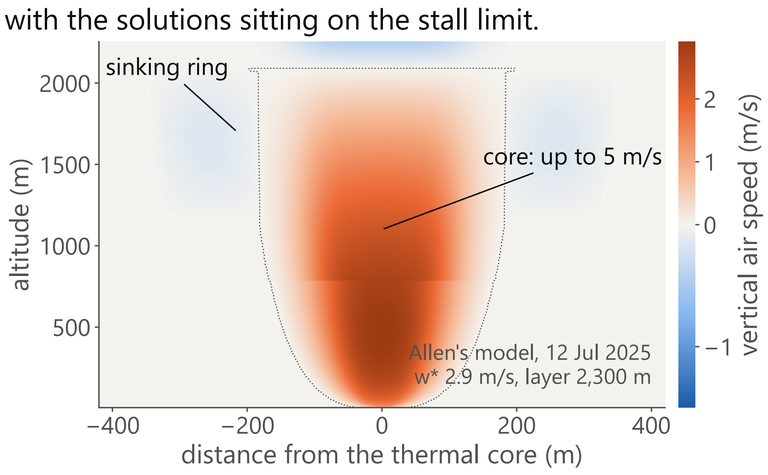
one DAgger-like round: retrained on the 1,095 states met in flight

1 The problem

A small motor-glider (SB-XC, **4.2 kg**) can save its battery by climbing in thermals, columns of rising air, like a glider pilot. But each thermal has a strong core, weak edges and sinking air around it, and it drifts with the wind.

- circle too wide: the aircraft misses the core;
- circle too tight: the inner wing stalls.

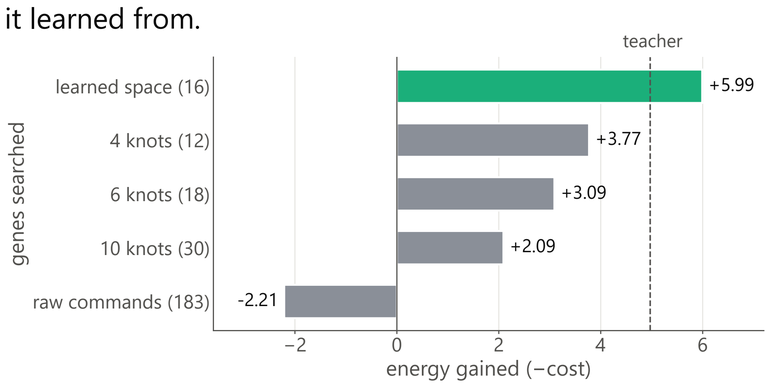
Planning the best circle is a hard, many-solution problem with the solutions sitting on the stall limit.



The thermal model used here (Allen, NASA 2006) on a real day: vertical air speed around the core.

2 Why a learned space?

The same genetic search, with the same time budget, does far better in the VAE's 16 learned numbers than in hand-made descriptions of the commands, and beats the teacher it learned from.



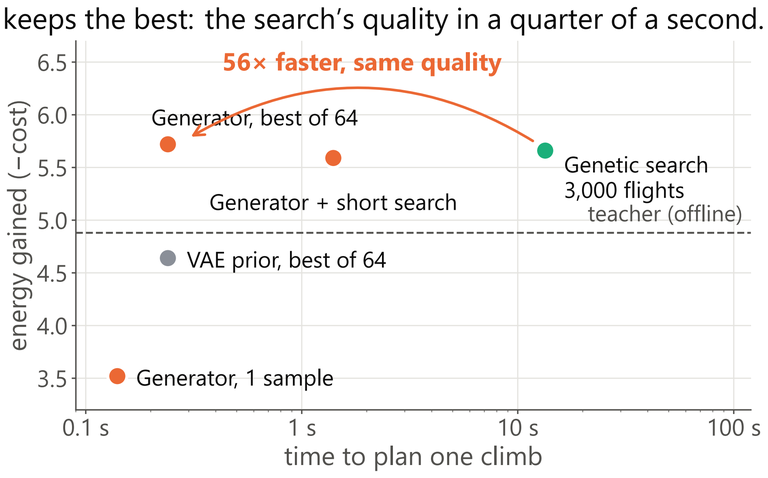
Energy gained in a 60 s climb (higher is better), 49 test climbs. Grey: commands described by knots, or raw.

3 Honest limits

- Simulation only; the aircraft's polar is estimated. With 20 % more mass, the search's lead shrinks.
- Given 2 minutes per climb, a nonlinear MPC beats the genetic search; given 20 s it solves half the climbs.
- On the windy day the blind generator lands out more often than the blind rule. Next: wind, then a flight test.

4 As good as search, 56× faster

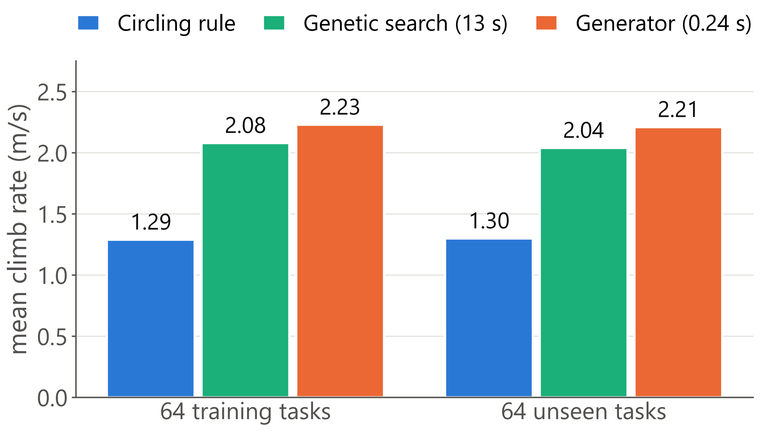
The search is good but slow. A generator trained on its answers proposes 64 climbs, flies each one in the simulator and keeps the best: the search's quality in a quarter of a second.



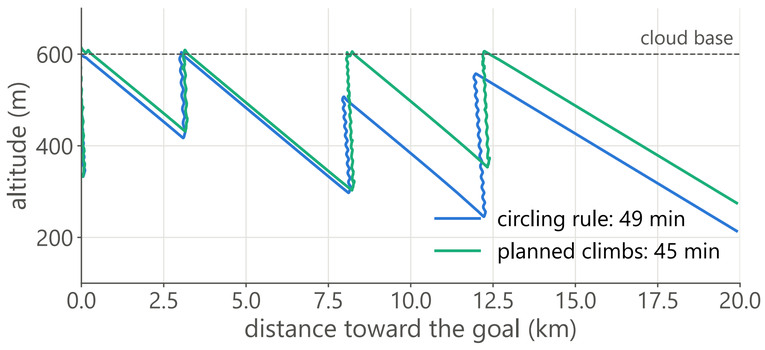
65 test climbs. One generator sample alone is not enough; 64 verified samples are.

5 In a 20 km cross-country flight

A tactician chooses which thermal to take; only the climbs differ. The generator climbs **1.7×** faster than the circling rule and keeps it on tasks it has never seen.



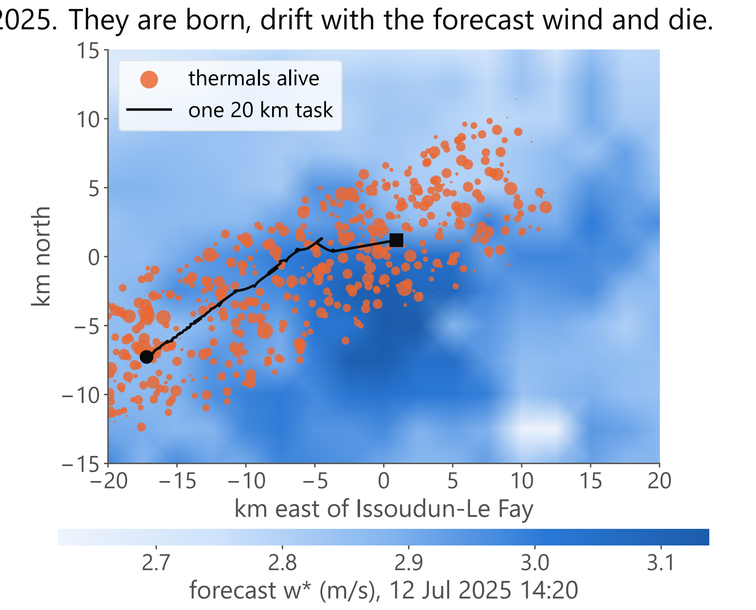
Mean climb rate, 64 tasks of 20 km each (MacCready 0.5). The DAgger-like round used the training tasks.



One task: faster climbs, higher glides, 4 minutes earlier, no battery used.

6 Into a real day

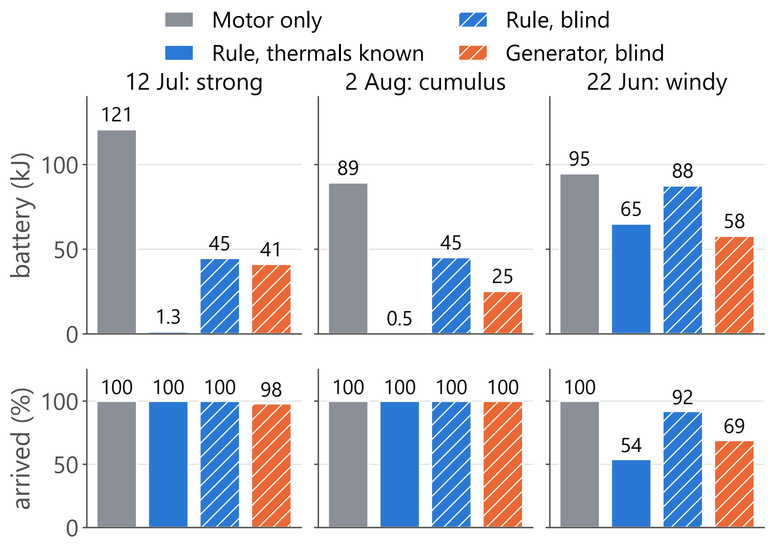
The thermals now come from a real weather forecast (DWD ICON-D2) over Issoudun, France, for three days of summer 2025. They are born, drift with the forecast wind and die.



12 July 2025: forecast thermal strength, the thermals alive at mid-flight and one task.

7 Flying blind

A real pilot does not see the thermal. The blind pilots feel it through the variometer, estimate its centre and strength with a Kalman filter (as ArduSoar does), and plan from that estimate.



48 tasks of 20 km per day, MacCready 1. Hatched: blind. Without seeing the thermals, the rule needs the motor to find them (45 kJ instead of 1). The generator, retrained on these weather thermals, needs **44 % less battery** than the blind rule on the cumulus day and 8 % less on the strong day. On the windy day it reaches the goal less often (69 % against 92 %).

Allen, Updraft model for autonomous soaring, NASA/TM-2006-214382 • Morando et al., flatness-based glide planning, ICRA 2025 • Lipman et al., Flow matching, ICLR 2023 • Ross et al., DAgger, AISTATS 2011 • Forecasts: DWD ICON-D2 via Open-Meteo (CC BY 4.0) • Computed on the Savio cluster (UC Berkeley).